

JEE MAIN REVISION MASTER: WAVE OPTICS SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 178 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Huygens' Principle & Interference (YDSE)	115 PYQs	Interference
Very High	Diffraction & Polarization (Brewster's & Malus' Law)	76 PYQs	Diffraction & Polarization

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Huygens' Principle & Interference (YDSE)	$\Delta x = \frac{y d}{D}$, $\beta = \frac{\lambda D}{d}$, $I_{\text{max}} = (I_1 + I_2)$, $\text{Shift due to slab} = \frac{(\mu - 1)t D}{d}$
Diffraction & Polarization (Brewster's & Malus' Law)	$a \sin \theta = n \lambda$ (minima), $\beta_0 = \frac{2 \lambda D}{a}$ (central max), $I = I_0 \cos^2 \theta$ (Malus), $\tan i_p = \mu$ (Brewster)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Huygens' Principle & Interference (YDSE) (Very High Yield — 115 PYQs)

Core Principle: Wavefront geometry, Young's double slit interference, fringe width $\beta = \lambda D/d$, and fringe shift on introducing thin mica sheet.

■ **Caution / JEE Pitfall:** Fringe width β depends on wavelength ($\beta \propto \lambda$). If YDSE is immersed in water ($\mu = 4/3$), fringe width decreases by factor of 4/3.

■ **Speed Trick:** Number of fringes shifted when a transparent slab of thickness t is placed: $N = ((\mu - 1)t / \lambda)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The maximum intensity in a Young's double slit experiment is I_0 . Distance between the slits (d) is 5λ , where λ is t ..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is..." [Ans: B]
- [2026 Morning Shift] Q#2 (MCQ): "In Young's double slit experiment, the fringe width of the interference pattern produced on the screen is $2.4\mu\text{m}$. If the experiment is ..." [Ans: B]

3.2 Diffraction & Polarization (Brewster's & Malus' Law) (Very High Yield — 76 PYQs)

Core Principle: Fraunhofer single-slit diffraction, width of central bright maximum, Brewster's angle, and Malus' intensity law for polarized light.

■ **Caution / JEE Pitfall:** In single slit diffraction, minima occur at $a \sin \theta = n\lambda$, whereas in YDSE interference, maxima occur at $d \sin \theta = n\lambda$!

■ **Speed Trick:** At Brewster's angle of incidence i_p , the reflected ray and refracted ray are mutually PERPENDICULAR ($i_p + r = 90^\circ$).

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of $5 \times$..." **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly p..." **[Ans: D]**
- **[2026 Morning Shift] Q#1 (Numerical):** "An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes ..." **[Ans: 30]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

The maximum intensity in a Young's double slit experiment is I_0 . Distance between the slits (d) is 5λ , where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $D=10d$ is _____.

- (A) $\frac{I_0}{4}$
- (B) $\frac{I_0}{2}$ **[Correct]**
- (C) I_0
- (D) $\frac{3I_0}{4}$

Correct Answer: B

Solution Analysis: A point exactly opposite to one of the slits is at a distance $y=(d/2)$ from the central maximum. The slit separation is given as $d=5\lambda$. The distance between slits and screen is $D=10d=50\lambda$. And the vertical distance of required pointy from the central point is $y=(d/2)=2.5\lambda$. For a point on the screen, path difference is approximately: $\Delta x=y \cdot (d/D)$. Substituting the given values: $\Delta x=((d/2)) \cdot (d/10d)=(d/20)$. Substituting $d=5\lambda$: $\Delta x=(5\lambda/20)=(\lambda/4)$. The relation between phase difference

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of 5×10^{-7} radian. The value of R is ____ cm .

- (A) 61
- (B) 122 **[Correct]**
- (C) 244
- (D) 305

Correct Answer: B

Solution Analysis: The ability of an optical instrument to distinguish between two closely spaced objects is defined by its resolving power. According to the Rayleigh criterion, for a circular aperture of diameter R , the minimum angular separation ($\Delta\theta$) that can be resolved is given by the formula : $\Delta\theta = (1.22\lambda / R)$ Where $\Delta\theta$ is the angular resolution in radians, λ is the wavelength of the incident light and R is the diameter of the objective lens or mirror. We are given the wavelength of light as $\lambda = 500\text{nm} = 500 \times 10^{-9}\text{m} = 5 \times 10^{-7}\text{m}$ And ang